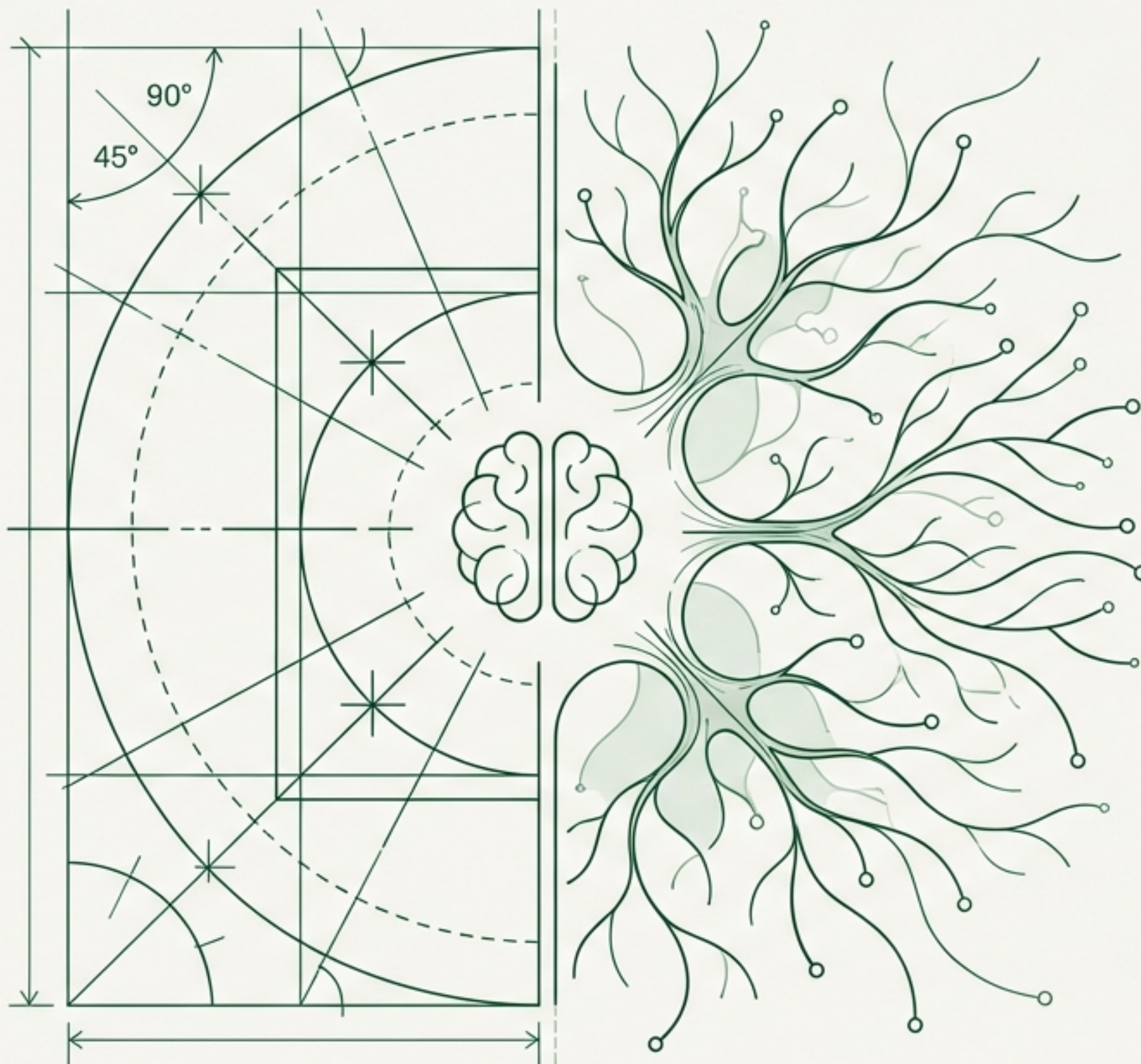




Verridian AI: A New Blueprint for AI Cognition

Moving beyond retrieval to build systems that understand.

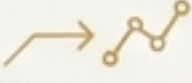
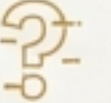


Key Focus: Structure + Understanding

The Filing Cabinet vs. The Brain

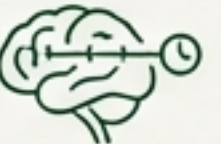
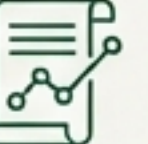


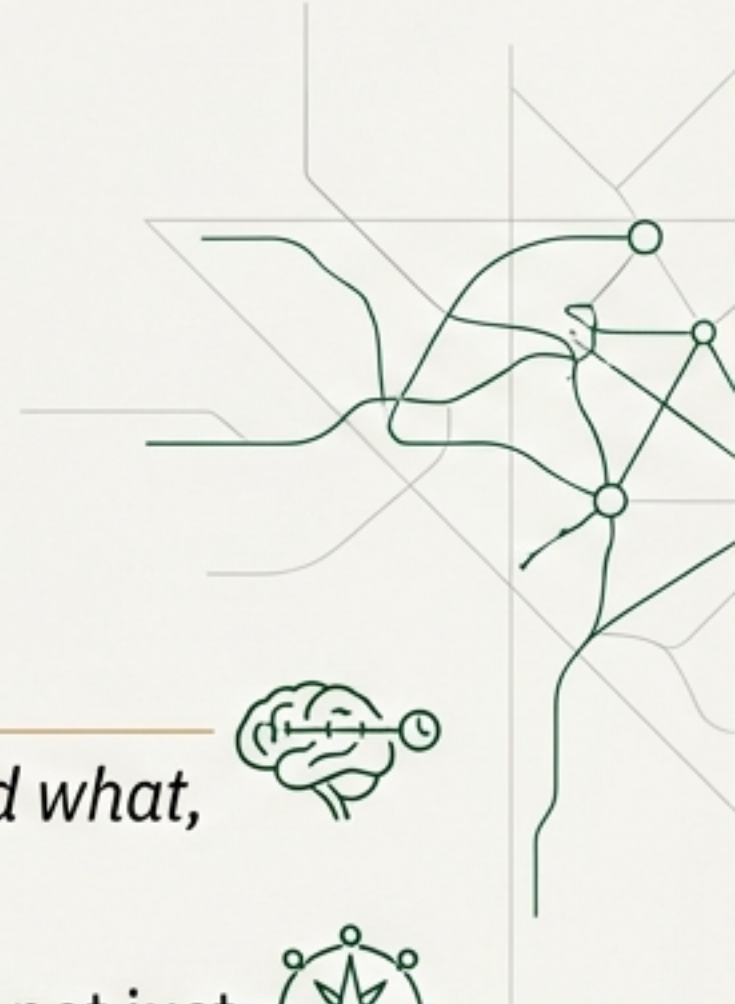
The Filing Cabinet

- ❑ Stores text as disconnected vectors. 
- ❑ Searches by semantic similarity. 
- ❑ Lacks context and understanding of relationships. 
- ❑ Prone to hallucination when context is ambiguous. 



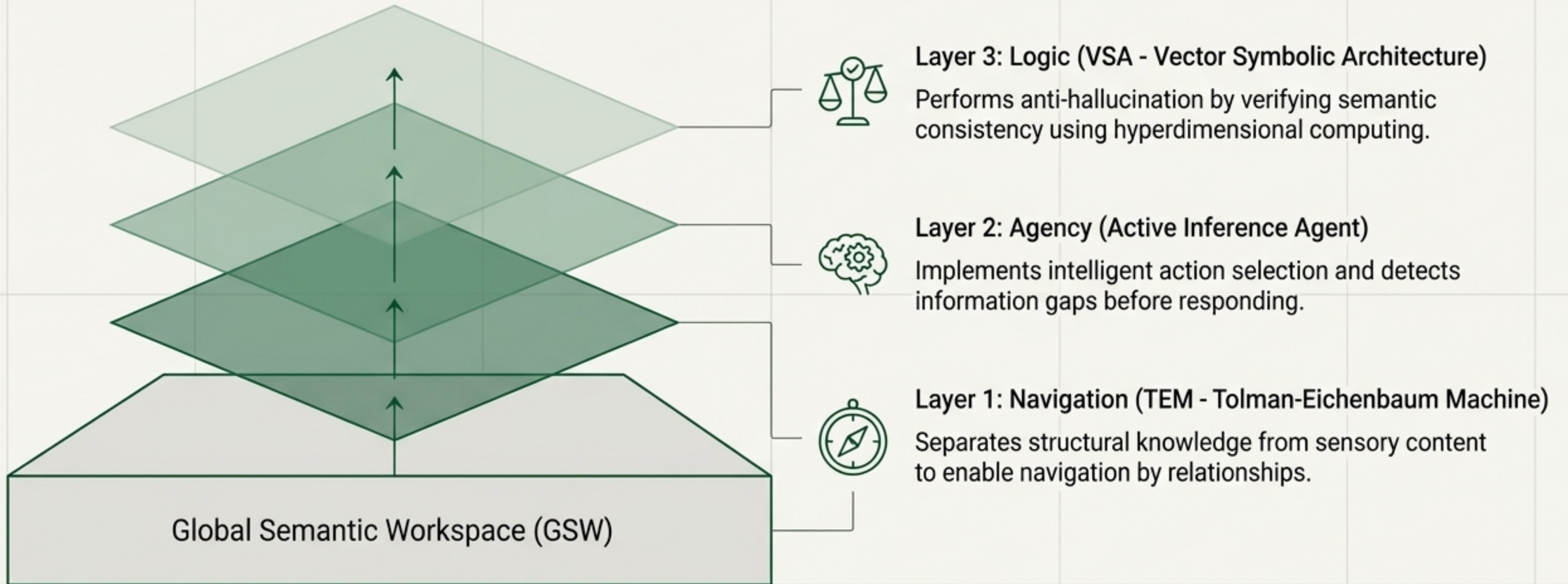
The Cognitive Brain

- Builds a memory of *who did what, when, and where*. 
- Navigates by relationships, not just similarity. 
- Understands the full narrative context. 
- Verifies logic to ensure truth. 



We are solving a fundamental architectural problem, not just making an incremental improvement.

Our Blueprint: A Three-Layer Cognitive Engine

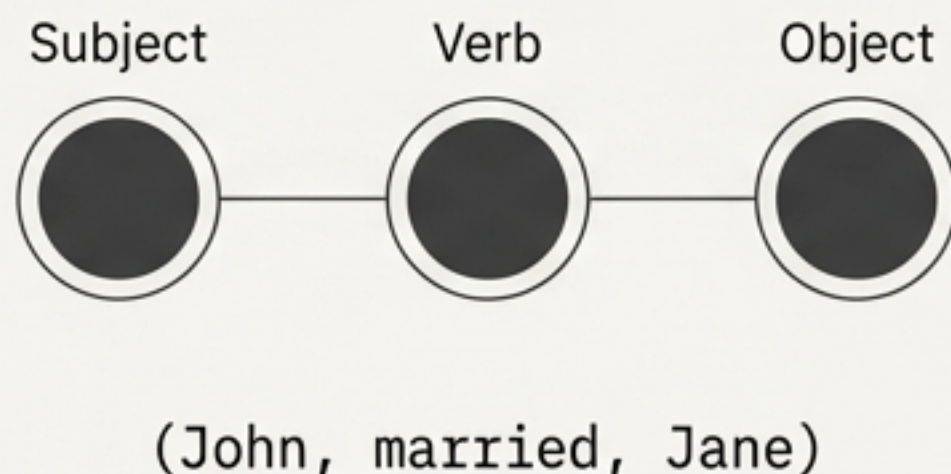


Our solution is a structured, brain-inspired architecture that mimics distinct cognitive functions: navigation, decision-making, and verification.

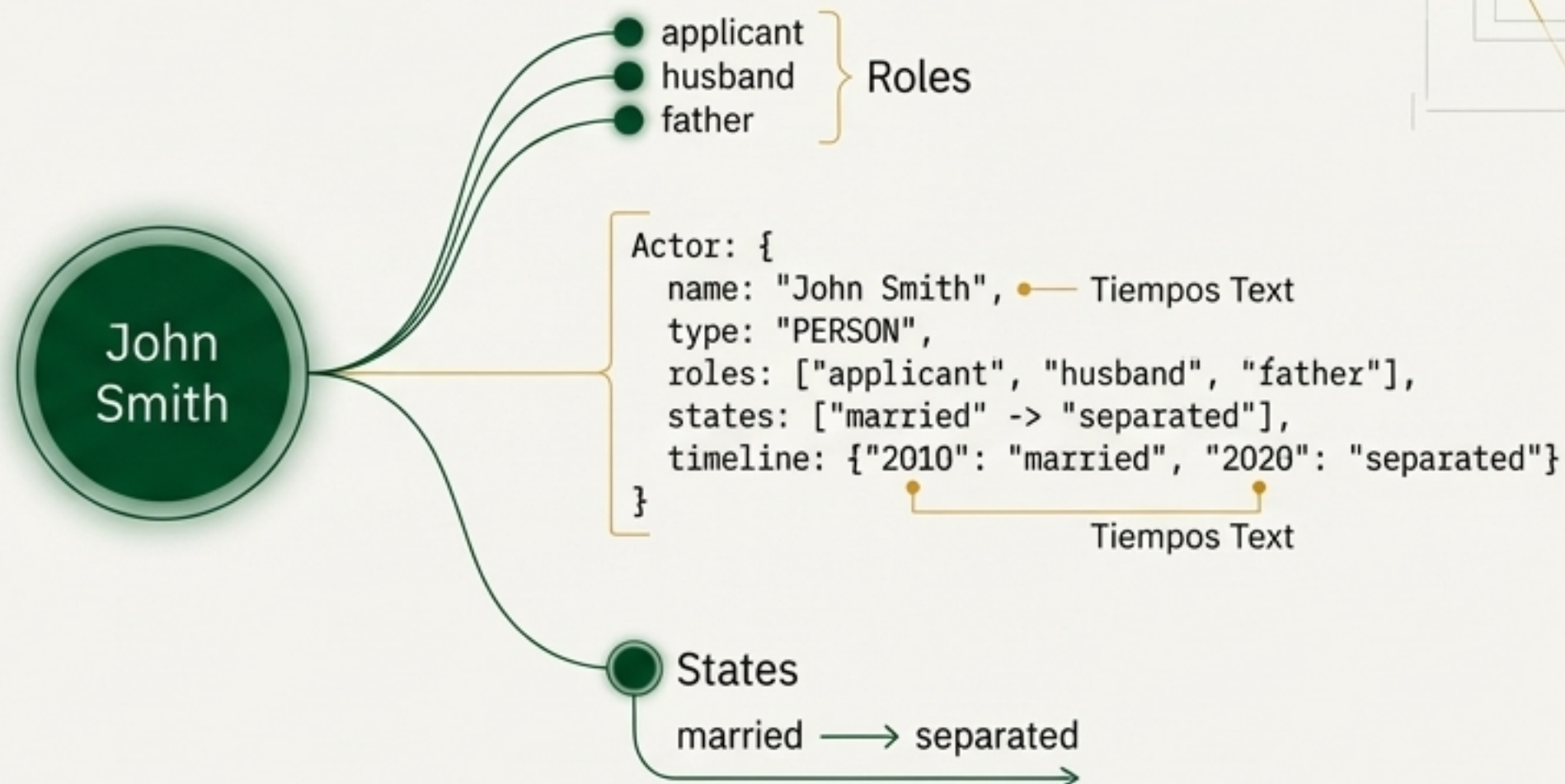
Pillar 1: Building Actor-Centric Memory

It remembers like a person, not a database.

Traditional Model

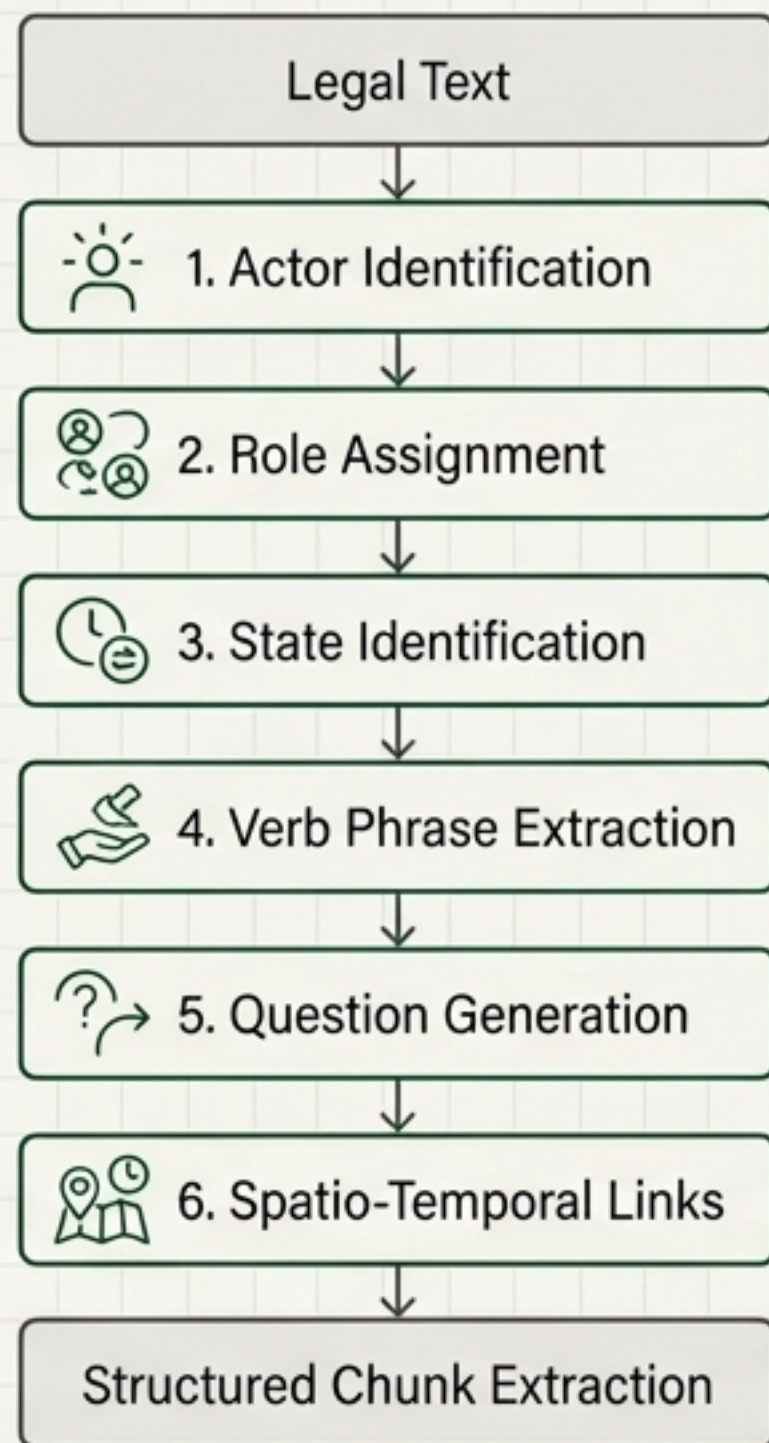


Actor-Centric Model



Moving from a 'bag of words' to a model of *who did what to whom, when, and where*.

How We Build the Memory: The 6-Task Extraction Pipeline



1. Actors: Identify central entities (people, assets, dates).

2. Roles: Assign legal functions (e.g., ‘applicant,’ ‘husband’).

3. States: Extract time-varying conditions (e.g., ‘MaritalStatus: separated’).

4. Verbs: Capture actions linking actors.

5. Questions: Generate predictive questions to identify gaps, mimicking episodic memory.

6. Links: Bind entities sharing the same time or location.

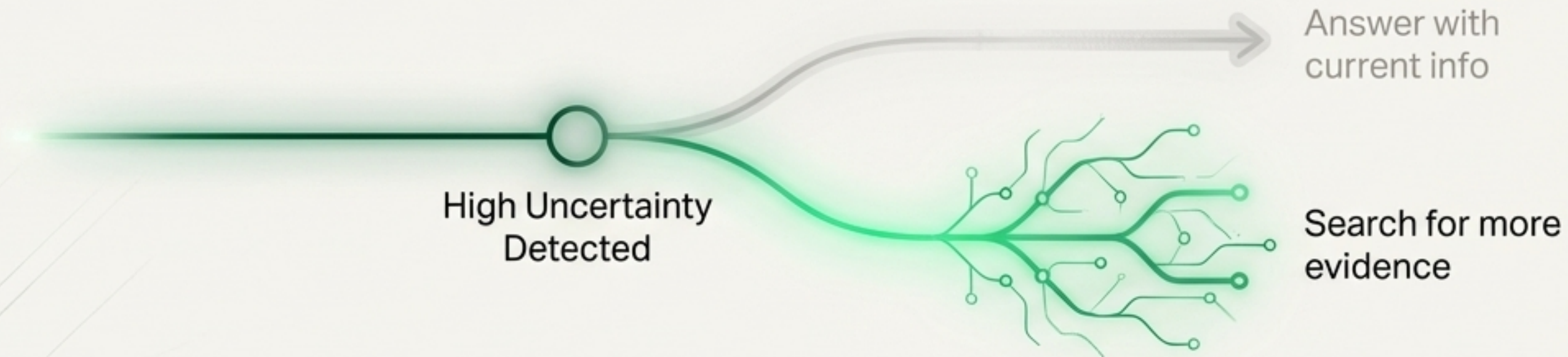
This is a rigorous, structured process for deconstructing language into a meaningful, actor-centric model.

Pillar 2: Intelligent Agency & Gap Detection

It actively hunts for missing information instead of guessing.

The system implements **Active Inference**, a framework to minimize 'Free Energy' (surprise or uncertainty). It constantly asks two cognitive questions:

- **Perception (VFE):** "What is happening?" — *Updating beliefs based on observations.*
- **Action (EFE):** "What should I do?" — *Selecting the action that will be most informative.*



Instead of hallucinating, our AI knows when its information is incomplete and takes action to resolve the uncertainty.

The Calculus of Curiosity: Expected Free Energy

Expected Free Energy (G) = Pragmatic Value + Epistemic Value



Pragmatic Value (Goal-Driven)

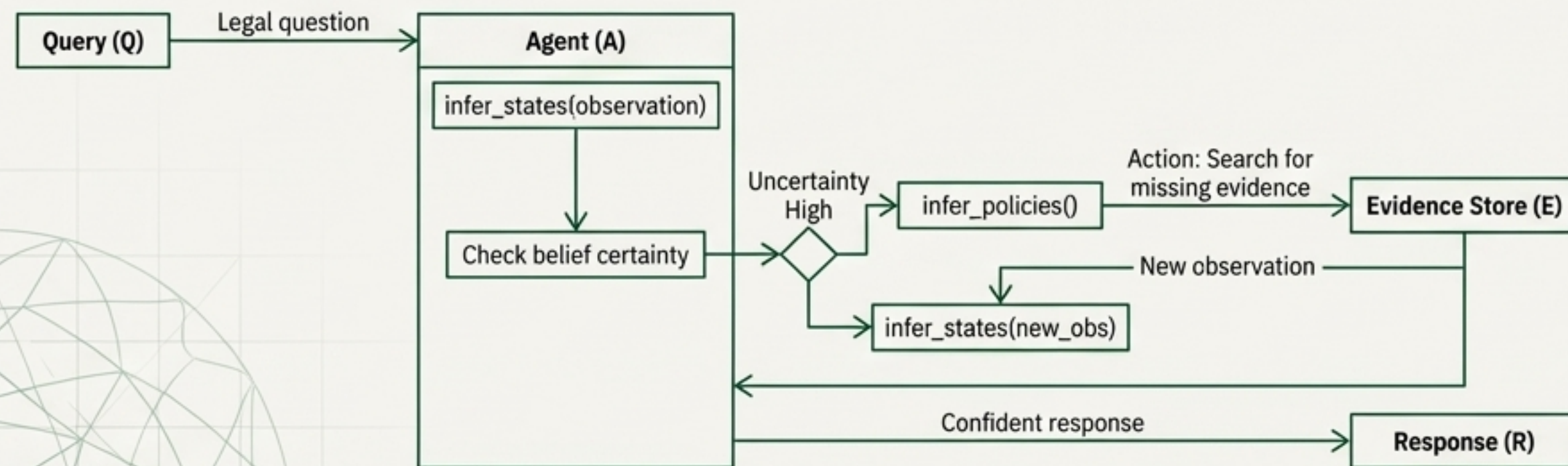
How useful is this action? Does it move me closer to my goal?



Epistemic Value (Information Gain)

How much will this action reduce my uncertainty? This is the system's curiosity.

Gap Detection Flow



Performance Stat: +3%
increase in accuracy and
reduction in hallucination
through epistemic exploration.

Pillar 3: A Logical 'Fact-Checker' for AI

Using hyperdimensional geometry to stop hallucinations.

The Vector Symbolic Architecture (VSA) is the system's Layer 3 verification engine. Concepts are represented as 10,000-dimensional vectors (hypervectors). Mathematical operations on these vectors are equivalent to symbolic reasoning. This allows the system to check for logical consistency, not just semantic similarity.



We don't just generate text; we verify its logical integrity against a set of rules before delivering an answer.

The VSA Triad: Binding, Bundling, and Permutation



Binding (Composition)

$$v1 * v2$$

Creates role-filler pairs (e.g., binding the vector for HUSBAND to the vector for JOHN). The result is dissimilar to both inputs.



Bundling (Superposition)

Element-wise addition + sign function

Creates sets or groups where the result is similar to all inputs (e.g., bundling all parties in a case).



Permutation (Sequence)

Cyclically shifting vector elements

Encodes sequence or order (e.g., the timeline married -> separated -> divorced).

Concrete Example

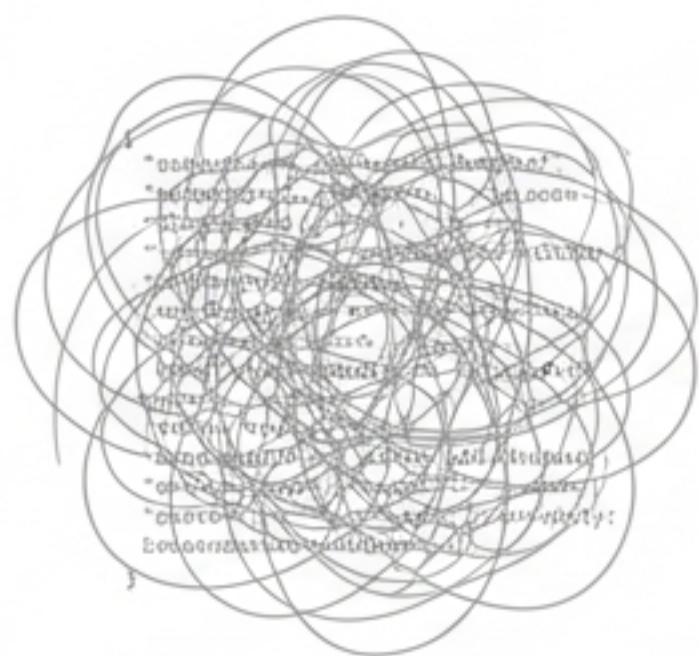
```
LOGIC_RULES = [  
    ("PROPERTY_SETTLEMENT", "REQUIRES", "MARRIAGE"),  
    ("SOLE_CUSTODY", "CONTRADICTS", "JOINT_CUSTODY"),  
]
```


Pillar 4: Engineering for Extreme Efficiency

It invented its own shorthand to maximize performance.

LLM context windows are the most expensive and slowest part of any AI system.

We engineered TOON (Token-Oriented Object Notation), a proprietary, compact serialization format designed specifically for LLM context injection.



VERBOSE JSON

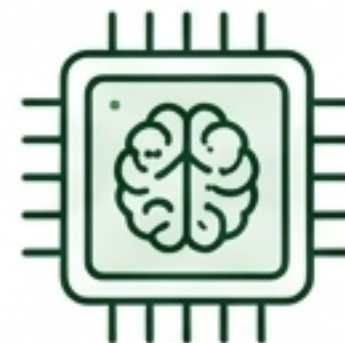


TOON



```
{*000211001111381011100  
30003031002131111021031  
02031303302322392032031  
00011512000000001041929  
11301203071132501200039  
000010002110232031011211  
20009230002012293531602  
20000900900217011031302  
00232132031232131021321  
1000331000000000001}}
```

TOON FORMAT



LLM CHIP

Deep architectural thinking extends all the way to the data format, ensuring our system is not only intelligent but also practical and efficient to operate at scale.

40% Token Reduction with Higher Accuracy

JSON - 45 tokens

```
{  
  "actors": [  
    {"id": "a1", "name": "John Smith",  
     "type": "person",  
     "type": "person",  
     "roles": ["Applicant", "Father"]} ]  
}
```

TOON - 27 tokens

```
Actors[1]{id,name,type,roles}  
a1,John Smith,person,Applicant|Father
```

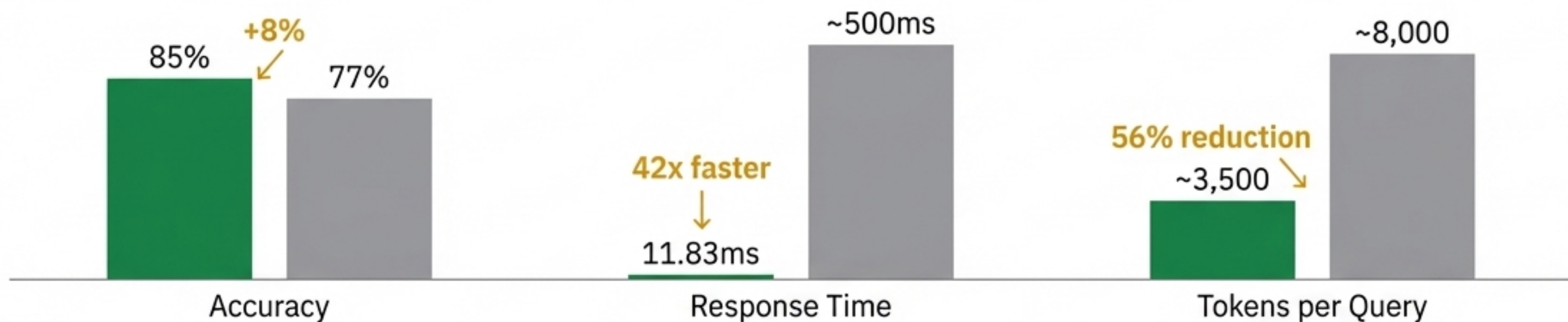
Entity Type	JSON Tokens	TOON Tokens	Reduction
Actor	45	27	40%
State	32	18	44%
VerbPhrase	38	22	42%
Overall Context	~8,000 (Trad. RAG)	~3,500 (Verridian)	56%

73.9%

Parsing Accuracy
(compared to 60%
for JSON).

The Results: Faster, Cheaper, More Accurate

Verridian vs Traditional RAG



Metric	Verridian	Traditional RAG	Improvement
Accuracy	85%	77%	+8%
Response Time	11.83ms	~500ms	42x faster
Tokens per Query	~3,500	~8,000	56% reduction
Query Success Rate	100%	~95%	+5%

This brain-inspired architecture delivers quantifiable, industry-leading performance across all key metrics.

Beyond Performance: A Commitment to Verifiable Trust

Our system is built for environments where accuracy is non-negotiable.
To ensure this, we've integrated advanced validation layers.

Statutory RAG



Every relevant claim is cross-referenced against an indexed corpus of legislation, including the *Family Law Act 1975* and *Child Support (Assessment) Act 1989*. This ensures all outputs are legally grounded.

Multi-Judge Evaluation

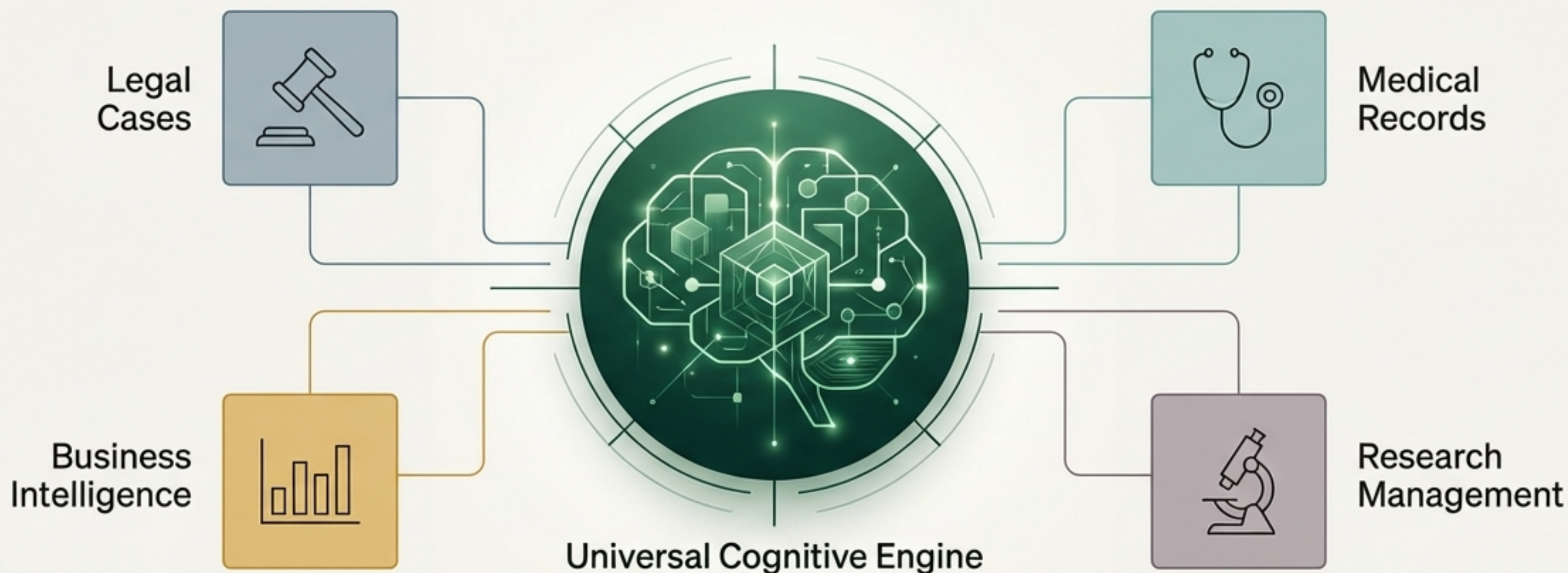


To mitigate single-model bias, responses are audited for quality, accuracy, and completeness by a panel of independent AI judges, with a consensus score calculated for final validation.

We don't just trust our internal logic; we verify it against external statutes and multi-model consensus.

The Architecture is Universal

We haven't just built a legal AI; we've built a cognitive blueprint applicable to any complex knowledge domain.



The core engine remains the same; only the entity types and relationships change.
This is a general-purpose architecture for structured understanding.

Building the Future of AI Cognition

The full research paper, 'Functional Structure of Episodic Memory', is available on arXiv.

arXiv: 2511.07587



Verridian-ai/Functional-Structure-of-Episodic-Memory

Contribute to the project on GitHub.

Learn more at **verridian.ai**



Created by Daniel Fleuren | Verridian AI